**Lina Alkarmi**
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## EDUCATION

**University of Michigan** — **Ann Arbor, MI**
*Doctor of Philosophy in Electrical and Computer Engineering* — *Expected December 2028 or May 2029*
*GPA: 4.00/4.00*
*Third-year PhD candidate. Advised by Professor Mingyan Liu.*

**George Mason University** — **Fairfax, VA**
*Bachelor of Science in Electrical Engineering* — *May 2024*
*GPA: 4.00/4.00*

## RESEARCH INTERESTS

*Mechanism Design:* Aligning individual incentives with collective welfare in AI-enabled sociotechnical systems.
*Strategic Machine Learning:* Understanding and designing for how people respond to algorithmic decision making.
*Security Economics:* Measuring the societal costs of cybersecurity failures to inform policy.

## SELECTED PUBLICATIONS

1. **Alkarmi, L.***, Chen, P.*, and Liu, M. (2026) Understanding Federated Learning Through the Lens of Mechanism Design: The Role of Data Heterogeneity, in International Conference on Game Theory and AI for Security, 1st ed. Springer Nature Switzerland, 2026, pp. 184–203. doi: 10.1007/978-3-032-40367-4_10. [* denotes equal contribution]
2. Huang, Z.*, **Alkarmi, L.***, and Liu, M. (2026) Multi-Level Strategic Classification: Incentivizing Improvement through Promotion and Relegation Dynamics, in Proceedings of the 43rd International Conference on Machine Learning, PMLR 306:46269-46308. [* denotes equal contribution]
3. **Alkarmi, L.**, Sarabi, A., and Liu, M. (2026) Estimating the Social Cost of Corporate Data Breaches, arXiv:2603.21270 (Appeared at WEIS 2026). Extended version under review at Journal of Cybersecurity.
4. **Alkarmi, L.**, Huang, Z., and Liu, M. (2025) When In Doubt, Abstain: The Impact of Abstention on Strategic Classification, in International Conference on Game Theory and AI for Security, 1st ed. Springer Nature Switzerland, 2025, pp. 124-144. doi: 10.1007/978-3-032-08067-7_7.

## PRESENTATIONS [underline denotes presenter]

1. **Alkarmi, L.**, Chen, P., and Liu, M. Understanding Federated Learning Through the Lens of Mechanism Design: The Role of Data Heterogeneity. Oral presentation at the 17th International Conference on Game Theory and AI for Security (GameSec), Ann Arbor, Michigan, October 26th, 2026.
2. Huang, Z., **Alkarmi, L.**, and Liu, M. Multi-Level Strategic Classification: Incentivizing Improvement through Promotion and Relegation Dynamics. Poster session at the International Conference on Machine Learning (ICML), Seoul, South Korea, July 9th, 2026.
3. **Alkarmi, L.**, Sarabi, A., and Liu, M. Estimating the Social Cost of Corporate Data Breaches. Oral presentation at the Workshop on the Economics of Information Security (WEIS), Berkeley, CA, June 2nd, 2026.
4. Huang, Z., **Alkarmi, L.**, and Liu, M. Sequential Strategic Classification with Multi-Stage Selective Classifiers. Poster session at the International Conference on Game Theory for Networks (GameNets), Evanston, IL, March 25th, 2026.
5. **Alkarmi, L.**, Huang, Z., and Liu, M. When In Doubt, Abstain: The Impact of Abstention on Strategic Classification. Oral presentation at the 16th International Conference on Game Theory and AI for Security (GameSec), Athens, Greece, October 14th, 2025.

## RESEARCH EXPERIENCE

**Liu Research Group** **Ann Arbor, MI**

*Research Assistant* *Aug 2024 to Present*

*Advisor: Mingyan Liu*

My research combines game-theoretic modeling with empirical measurement to understand how AI-driven and institutional systems shape human behavior, and how to design them toward goals such as welfare and fairness.

***Understanding Federated Learning Through Mechanism Design (GameSec 2026)***

- Addressed free riding in federated learning with heterogeneous data by adapting a classical public goods mechanism (the Externality mechanism), and compared it with a recently proposed reciprocal-fairness mechanism (Shapley).
- Showed that the Shapley mechanism can lead agents to over-contribute data and fail to guarantee that participating beats training alone, while the Externality mechanism avoids both problems but does not guarantee fairness.
- Results indicate when each applies: Shapley when fairness matters most, Externality when collective benefit and making every agent better off matter more.

***Multi-Level Strategic Classification (ICML 2026)***

- Introduced a multi-level framework in which agents progress through a sequence of classifiers (promotion on passing, relegation on failing, and a hold on an inconclusive result) and can mix gaming and improvement at every step.
- Modeled the agent's long-term problem as a discounted Markov decision process with farsighted agents and skill retention, and characterized the agent's optimal policy.
- Proved conditions under which never gaming is the optimal long-term strategy even though gaming is cheaper at every single step, so that improvement alone carries agents to any target level.
- Derived design guidelines (number of levels, spacing, difficulty) and an algorithm that builds a classifier sequence for a target level, for settings such as education, lending, and hiring; released the algorithm and code on GitHub.

***Estimating the Social Cost of Corporate Data Breaches (WEIS 2026, under review at JCS)***

- Defined the social cost of a data breach from the victim's side (out-of-pocket losses, opportunity costs, healthcare expenditures) and developed upper and lower bound calculation methods for it.
- Linked thirteen years of national victimization survey data (2008 to 2021) to breach records and found a statistically significant rise in identity theft after mega-breaches, with a 1-to-2-month discovery lag. Released pipeline on GitHub.
- Applied the framework to case studies such as the Heartland 2009 breach and the Target 2013 breach, finding that settlements underestimated social cost by factors of 5 to 16.

***When In Doubt, Abstain: Abstention in Strategic Classification (GameSec 2025)***

- Investigated how a classifier's option to abstain (deferring to additional review when uncertainty is high) affects the strategic behavior of self-interested agents.
- Proved that abstention never hurts the principal's loss and can deter manipulation, making gaming costlier for less-qualified agents, giving decision makers principled guidance on how much to abstain.

**MIT Uncertainty Quantification Group** **Cambridge, MA**

*Research Assistant* *Jun 2023 to Dec 2023*

*Advisor: Youssef Marzouk*

- Developed an efficient nonlinear Bayesian filtering method for chaotic systems.
- Computed the SRB measure of a dynamical system and sampled using Langevin dynamics to reconstruct the density.
- Project yielded a new filtering method applicable to high-dimensional systems to accelerate density computations.

**GMU SatCom/CubeSat Lab** **Fairfax, VA**

*Research Assistant* *Jun 2022 to Jun 2023*

*Advisor: Peter Pachowicz*

- Designed a high-precision laser controller to serve as an artificial star reference for telescope calibration, intended for spaceflight on a NASA funded mission.
- Reviewed technical specifications and device manuals to inform controller design requirements.
- Designed the controller PCB in KiCAD, managed fabrication, and assembled the board through soldering.
- Conducted physical experiments to characterize the thermal performance of the TEC cooling system, collecting temperature data across multiple current levels.

## PROFESSIONAL EXPERIENCE

**Parsons (Blackhorse Solutions)** **Herndon, VA**

*Signal Processing Intern* *Jun 2024 to Aug 2024*

- Focused on message parsing and satellite position calculations for GNSS signals (GPS and other constellations).
- Developed C++ code to parse GNSS messages and calculate almanac and ephemeris satellite positions within 50 km and 50 m, respectively.
- Developed a visualization script in Octave to plot satellite position with respect to the Earth.

**ManTech** **Chantilly, VA**

*Forensics Engineering and Development Intern (Seasonal)* *Jun 2020 to Aug 2022*

- Engineered an end-to-end recruitment application in Python with a multi-class NLP model achieving 83.75% accuracy, deployed for internal beta testing and directly influencing a company-wide automated recruitment pipeline.
- Reverse-engineered proprietary software using Ghidra and x64dbg to identify unique digital signatures within media files.
- Performed static analysis on APK files using YARA signatures to extract metadata and identify security vulnerabilities.

## FELLOWSHIPS AND AWARDS

**Parsons 2024 Top Performing Intern Award** (Awarded Aug 2024): Recognized as one of two Parsons cyber and intelligence 2024 top performing interns.

**GMU Outstanding Academic Achievement Award** (Awarded May 2024): Departmental award recognizing academic excellence, ECE Department, George Mason University.

**National Science Foundation Graduate Research Fellowship** (Awarded April 2024): Premier national fellowship for STEM research.

**Goldwater Scholarship** (Awarded March 2023): Premier national undergraduate scholarship for research excellence in STEM.

**University Scholar** (Awarded March 2020): Full tuition merit-based scholarship at GMU awarded to less than 0.2% of applicants to the Honors College.

**Bioengineering Research Fellow** (Awarded March 2020): Competitive research fellowship offered to outstanding incoming University Scholars for early research involvement in Bioengineering at GMU.

**Prudential Spirit of Community Distinguished Finalist** (February 2020): National volunteer recognition program; selected as Distinguished Finalist for co-founding Princess Packages, a nonprofit serving hospitalized and homeless children.

## TEACHING EXPERIENCE

**George Mason University** **Fairfax, VA**

*Learning Assistant, ECE 321: Continuous Time Signals and Systems* *Jan 2023 to May 2023*

- Held two weekly office hour sessions, assisted students with homework, and answered questions.
- Created study guide for students and held review sessions prior to exams.

## SERVICE AND LEADERSHIP

**Peer Reviewer**: ICML 2026, NeurIPS 2026, GameSec 2026.

**University of Michigan** **Ann Arbor, MI**

*Graduate Student Engineering Ambassador (GSEA)* *May 2026 to Present*

- Coordinate outreach events and attend undergraduate research seminars as a mentor.
- Advise prospective and current students individually on PhD life, research, and academic culture at Michigan Engineering.
- Organized an outing for 40 engineering students to the Michigan vs. Penn State men's soccer game (Oct 2026), including ticket purchasing, sign-ups, and email coordination.

**University of Michigan ECE Department** **Ann Arbor, MI**

*Student Volunteer* *Sep 2024 to Present*

- BudEE mentor: Mentor a first-year ECE student (Sep 2026 to Present).
- Planned the ECE department Halloween party: built the budget, ordered supplies, booked the venue and tables, recruited volunteers through a sign-up system, and ran publicity (Oct 2026).
- L4DC 2025 student volunteer (Jun 2025).
- NSF GRFP Panelist**:** Advised prospective applicants on the Graduate Research Fellowship Program (Sep 2024).

**George Mason University Honors College Alumni Network** **Fairfax, VA**

*Alumni Mentor* *Aug 2026 to Present*

- Mentor an undergraduate student applying to graduate and PhD programs (meeting on Zoom).
- Conduct one-on-one virtual interviews with current GMU students.

**Princess Packages** **Sterling, VA**

*Co-Founder and Director* *Nov 2017 to 2023*

- Co-founded a 501(c)3 nonprofit that gathers volunteers to create handmade care packages for hospitalized and homeless children.
- Organized large volunteer events (100+ people), secured grant funding (2018 Step Up Loudoun Youth grant, 2019 Forward Turn grant), and served as liaison between hospitals, homeless shelters, and the organization.
- Expanded the organization nationally and designed website, fundraisers, and outreach materials.

## SKILLS

*Research Methods:* Stackelberg games, Markov decision processes (dynamic programming), equilibrium and incentive analysis, theoretical proofs, survey-weighted statistics, hypothesis testing, linking datasets across sources

*Programming/Tools:* Python (NumPy, Scikit-Learn, Pandas, Matplotlib, SciPy, Statsmodels), MATLAB/Octave, Shell (Bash), Git, Linux/Unix, C++

*Machine Learning:* TensorFlow, PyTorch, reinforcement learning, Bayesian filtering